

Priliminary Algorithm for Horn Classification.

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1 Priliminary Classification

We record audio 1 second chunks of audio data and pass it through a CNN classifier network. A pretrained classifier trained on youtube data known as YAMNet is used for this perpose. This classifies the data into 512 different chunks, and returns an encoding vector, which is later used for further classifying horndata.

The YAMNet network has the following classes which are relevant for us:

- Vehicle
- Vehicle Horn
- Car Horn
- Air Horn, Truck Horn
- Bicycle Bell
- Honk

The model is simple enough that we maybe able to transmit this data over the wire to a centralized server for further processing.

2 Principle Component Analysis and Clustering

To further improve the classification granularity we add a PCA layer which effectively reduces the irrelevant dimentions from the 1024 dimentional data. We also perform extractions on label by first sorting the probabiltiy vector in descending order and picking the first 'n' elements whose sum is greater than 95%.

After this processing we pass the PCA embeddings to an real-time classifier calle Birch. This clusters together all the different kinds of horns into different section. We can ensure we only classify horns by limiting our search to the previously mentioned classes.

3 Labeling Data by using LLMs

To label the clusters created by Birch we utilize an LLM such as ollama. We pass the labels that have been collected through the duration of data collection. This allows the LLMs to predict which class the audio could belong to. Effectively giving us the vehicle where the horn originated. The classes Car/Truck are present, motorbike is not however a Motorcycle noise is a classifier category in YAM thus the LLM may be able to extract this information out of the system.

We are running the ollama LLM locally on the machine to avoid having to connect to a resource expensive network connection.

4 Problems present in current method

There are a few issues in the current method.

- The Audio data is highly dependent on the recording system and the environment.
- The LLMs (locally run) tend to hallucinate data, thus sometimes misnaming some categories.
- Unstructured output from LLMs
- The Clustering has to happen at higher than 5dims to be effective.

However despite these issues, The YAM network provides us with enough flexibility that we may be able to proceed.

5 Graphs after training.

This is an example of this algorithm at work. I have not enabled the filtering of different audio to check if the LLM worked correctly, and unfortunately due to the small parameter count it hallucinated a lot.

[464] :	X	Y	catagory	label	processed_labels
0	1.336193	1.322242	0 Vehicle; Reversing beeps; Squeal; Truck; Anima...		car alarm
1	-1.178361	1.598715	1 Vehicle; Engine; Sound effect; Motor vehicle (...)		motorboat
2	2.642228	0.016636	2 Speech; Radio; Outside, urban or manmade; Vehi...		alarm
3	0.236208	-1.547206	3 Speech; Outside, urban or manmade; Inside, lar...		speech
4	0.287742	-2.095090	3 Speech; Telephone; Inside, small room; Narrati...		speech
5	2.110316	-0.307875	2 Speech; Inside, small room; Alarm; Buzzer; Ala...		alarm
6	-2.879106	-0.439554	4 Skidding; Vehicle; Motor vehicle (road); Speec...		Tire squeal
7	1.309452	0.081285	2 Speech; Vehicle; Truck; Outside, urban or manm...		alarm
8	0.727360	-1.957845	3 Speech; Outside, urban or manmade; Inside, lar...		speech
9	0.406555	0.205694	5 Speech; Engine; Motor vehicle (road); Vehicle;...		idling
10	-1.328828	-0.518338	6 Speech; Engine; Vehicle; Engine starting; Idli...		engine
11	0.643401	-0.993273	3 Speech; Narration, monologue		speech
12	-0.294921	-0.597480	6 Speech; Television; Vehicle; Motorboat, speedb...		engine
13	-2.935417	0.063372	4 Sound effect; Horse; Neigh, whinny; Music; Liv...		Tire squeal
14	-1.509744	1.061289	1 Vehicle; Motor vehicle (road); Car; Engine; Tr...		motorboat
15	-2.353665	2.216445	1 Vehicle; Gurgling; Water; Boat, Water vehicle;...		motorboat
16	1.135480	0.780508	0 Speech; Water; Stream; Vehicle; Gurgling; Radi...		car alarm
17	0.021357	-0.942440	3 Speech; Inside, small room; Music; Child speec...		speech
18	-1.989049	-0.915099	4 Speech; Laughter; Horse; Clip-clop; Animal; Sn...		Tire squeal
19	2.426170	0.218756	4 Speech; Music; Vehicle; Outside, urban or manm...		Tire squeal

Figure 1: Example of classification of generated vector.

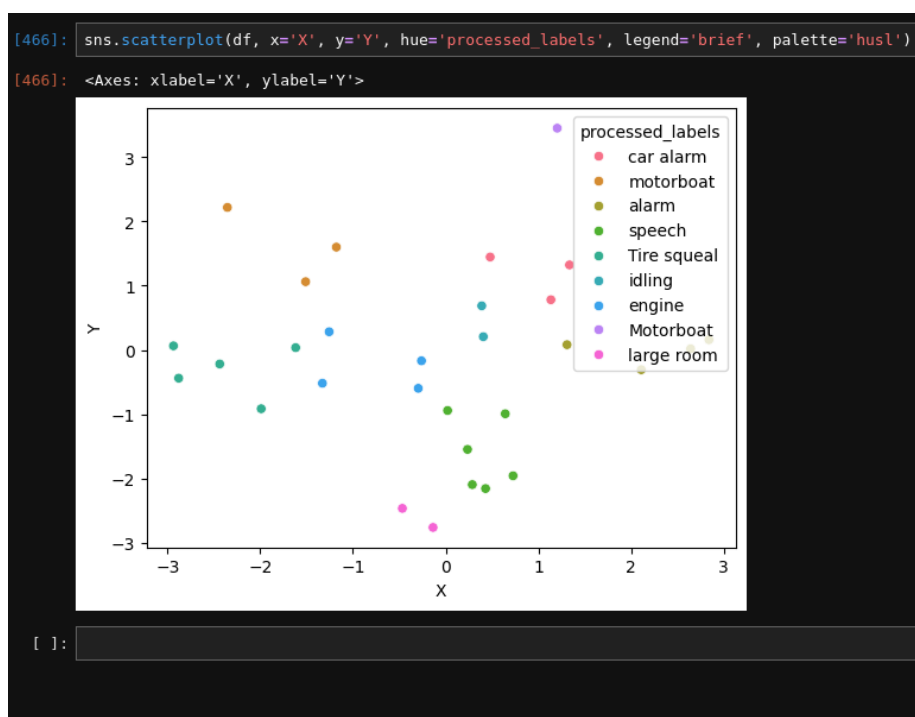


Figure 2: After Clustering graph.